FUNÇÕES TRIGONOMÉTRICAS

$\sin\theta = \frac{CO}{H}$

$\cos\theta = \frac{CA}{H}$

$\operatorname{tg}\theta = \frac{CO}{CA}$

$180° \to \pi \text{ rad}$

$\left[\frac{180°}{\pi} = \frac{\theta°}{\theta}\right]$

$\left[\frac{\pi}{180°} = \frac{\theta}{\theta°}\right]$

$\sin\theta$

$\cos\theta$

	30°	45°	60°
$\sin\theta$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\operatorname{tg}\theta$	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

$180°$

$0 - 90°$

$90° - 180°$

FUNÇÃO SENO $f: \mathbb{R} \to [-1,1]$ $f(x) = \sin x$ $T = 2\pi$ $f(x+2\pi)$ $I = \mathbb{R}$

FUNÇÃO COSSENO $f: \mathbb{R} \to [-1,1]$ $f(x) = \cos(x)$ $T = 2\pi$ $f(x+2\pi)$

FUNÇÃO TANGENTE $f(x) = \operatorname{tg}(x)$ $\operatorname{tg}(x) = \frac{\sin(x)}{\cos(x)}$, $\forall \cos(x) \neq 0$

FUNÇÃO COTANGENTE $f(x) = \cot g(x)$ $\cot g(x) = \frac{\cos(x)}{\sin(x)}$, $\forall \sin(x) \neq 0$ $\operatorname{Im} \neq \pm\infty$

FUNÇÃO SECANTE $f(x) = \sec(x) = \frac{1}{\cos(x)}$ $\forall \cos(x) \neq 0$ $\operatorname{Im} \neq \pm 1$ $\left[\frac{\sin(x)}{\cos(x)}\right]$

FUNÇÃO COSSECANTE $f(x) = \operatorname{cosec}(x) = \frac{1}{\sin(x)}$ $\forall \sin(x) \neq 0$ $\operatorname{Im} \neq \pm 1$

FUNÇÕES TRIGONOMÉTRICAS INVERSAS

FUNÇÃO ARCO SENO $f^{-1}: [-1,1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$ $f(x) = \arcsin(x)$

FUNÇÃO ARCO COSSENO $f^{-1}: [-1,1] \to [0,\pi]$ $f^{-1}(x) = \arccos(x)$

FUNÇÃO ARCO TANGENTE $f^{-1}: \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ $f^{-1}(x) = \operatorname{arctg}(x)$

$f(x) = \sin x$

$f(x) = \sin(x) = 1$

$f^{-1}(x) = \sin x = 1$

$\frac{\pi}{2}$

EXM 1) $f(x) = 3 \cdot \cos\left(x - \frac{\pi}{2}\right)$

$f(x) = \cos(x)$

$f(x) = 3\cos(x)$

$f(x) = 3 \cdot \cos\left(x - \frac{\pi}{2}\right)$

$x = \frac{\pi}{2}$ $f(\frac{\pi}{2}) = 3 \cdot \cos\left(\frac{\pi}{2} - \frac{\pi}{2}\right) = 3 \cdot \cos(0) = 3 \cdot 1 = 3$

$T = 2\pi$

$T = \frac{2\pi}{1}$

$T = \frac{2\pi}{1}$

$T = \frac{2\pi}{1}$

$T = \frac{2\pi}{1}$

$f(x) = \cos(2x)$

$x = 0$ $\cos 0 = 1$

$x = \frac{\pi}{2}$ $\cos\left(2 \cdot \frac{\pi}{2}\right) = \cos \pi = -1$

$x = \frac{\pi}{4}$ $\cos\left(2 \cdot \frac{\pi}{4}\right) = \cos \frac{\pi}{2} = 0$

$x = \pi$ $\cos(2 \cdot \pi) = 1$

$T = \pi$

EXM 2) $\sin(x) = \frac{\sqrt{3}}{2}$ $\operatorname{Im} = \{y \in \mathbb{R} / -1 \leq y \leq 1\}$

$-1 \leq \frac{\sqrt{3}}{2} \leq 1$

$-1 \leq 0{,}866 \leq 1$

$-1 \leq \sin x \leq 1$

$\boxed{-1 \leq x \leq 1}$

EXM $y = \sin\left(\frac{\pi}{3}\right) \cdot 3 \cdot \operatorname{tg} 14 + \sin\left(-\frac{\pi}{3}\right)$ $\sin\left(-\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right)$

$-\frac{\pi}{3} \leq \frac{\pi}{3} \leq \frac{\pi}{3}$

$\cos\alpha = \cos\beta$

$\frac{\pi}{3} = -\cos\beta$

$\cos(-\beta) = \cos\beta$

$\cos(-\beta) = \cos\beta$

$\operatorname{tg}(x) = \frac{\sin x}{\cos x}$

$\cos(-\beta) = \cos\beta = \cos 2\pi - \cos\left(\frac{\pi}{3}\right)$

$y = \sin\left(\frac{\pi}{3}\right) \cdot 3 \cdot \operatorname{tg}(14) + \sin\left(-\frac{\pi}{3}\right) \Rightarrow y = 0$

EXM 3) $\frac{\sin(x)}{\cos x} = \frac{\sin x \cdot \sin x}{\cos x}$ $\sin^2 = \frac{1}{\sin x}$

$\frac{\sin x \cdot \sin x}{\cos x} = \frac{\sin x \cdot \sin x}{\sin x}$ $\cos \sin x = \frac{1}{\sin x}$

$\sin x = \frac{\sin x}{\sin x} = \sin x \cdot \frac{\sin x}{\sin x} = 1$

$\cos^2 x + \sin^2 x = 1$

$r = 1$

$\boxed{\cos^2 x + \sin^2 x = 1}$

EXM $\sin(x) = \frac{3}{5}$ $\frac{\pi}{2} < x < \pi$

$\cos(x) = -\frac{4}{5}$

$\operatorname{tg} x = ?$

$\sin \theta = 0$

$\cos \theta = 1$

$\operatorname{tg} x = \frac{\sin x \cdot \sin x}{\cos x} = -\frac{3}{5} \cdot \frac{5}{-4} = \frac{3}{4}$

Revisão

01 $f(x) = 2^x$ $f(x+1) = 2^{x+1}$

$f(x+2) = 2^{x+2}$

$2^{x+1} = 2^x + 2^{x+2} = \frac{2^x}{2^x} = \frac{2^x}{2^x} + \frac{2^x}{2^x} \cdot 2^2$

$2^x(2-1) = 2^x \cdot 2^2$

$\frac{2^x}{2^x} = \frac{2^x \cdot 2^2}{2^x}$

$\frac{2}{2} = \frac{2^2}{2}$

Domínio

$f(x) = \sqrt{\frac{x}{x-3} - \frac{x}{(x+3)}}$

$\frac{x}{x-3} - \frac{x}{(x+3)} \geq 0$

$\theta = x^2 + 3x$

$x^2 + 3$

$x = \pm 3$

$\boxed{\text{Cria } x_1 = -3}$ $\boxed{\text{Cria } x_2 = 3}$ $\boxed{\text{Cria } x_3 = -3}$

$\frac{x}{x-3} - \frac{x}{(x+3)} > 0$ $\frac{x}{x-3} - \frac{x}{(x-3)} > 0$ $\frac{x}{x-3} - \frac{x}{(x+3)} > 0$

EXERCÍCIOS 4 A 10 DAS ATIVIDADES ANTERIORES (PÁG 04 - PÁG 07)

EXM 1) $f(x) = x^2 - 2x - 2x$

$f(x) = x^2 - 2x - 2x$

$f(x) = x^2 - 2x - 2x + 2 - 2$

$f(x) = (x^2 - 2x + 1) - 2 - 2$

$f(x) = -(-x^2 + 2x + 1) - 2$

$f(x) = -(-x^2 + 2x + 1) - 2$

$f(x) = -(x-1)^2 - 2$

$(x-1)^2 = x^2 - 2x + 1$

$f(x) = |x+1|^2$

$f(x) = |x+1|^2 - 2$

$f(x+1) = (x+1+1)^2 = 0$

$f(x+1) = (x+1+1)^2 - 2 = 0 - 2 = -2$

$f(x) = -(x-1)^2 - 2 = -(1-1)^2 - 2$

$f(x) = -(x-1)^2 - 2$

EXM 2) $f(x) = \frac{x}{\sqrt{1+x}}$ $\begin{cases} f(x) \\ f'(x) \end{cases}$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$f(x+h) = \frac{x}{\sqrt{1+(x+h)}}$ $f(x) = \frac{x}{\sqrt{1+x}}$

$f(x+h) = \frac{x}{\sqrt{1+x+h}}$

$f(x+h) = \frac{x}{\sqrt{1+x+h}} - \frac{x}{\sqrt{1+x}}$

$\frac{1}{(f)^2} = \frac{1}{\frac{x}{\sqrt{1+x}}} - \frac{x}{\sqrt{1+x}} = \frac{\sqrt{1+x}}{x} - \frac{x}{\sqrt{1+x}} = \frac{\sqrt{1+x}}{x} - \frac{x}{\sqrt{1+x}}$

$\frac{1}{f} = \frac{1}{\sqrt{1+x}}$ $\frac{1}{f} = \frac{x+1-x}{\sqrt{1+x}} = \frac{x+1}{\sqrt{1+x}}$

$f \frac{(x+h)}{h} - \frac{f(x)}{h} = \frac{1}{(\sqrt{1+x+h} + \sqrt{1+x})} - \frac{x}{h(\sqrt{1+x+h} + \sqrt{1+x})}$ $\frac{1}{f} = \frac{\sqrt{1+x}}{x}$

$\frac{1}{f} = \frac{\sqrt{1+x}}{x}$

$\frac{1}{f} = \frac{\sqrt{1+x}}{x}$

$\frac{1}{f} = \frac{\sqrt{1+x}}{x}$

EXM 3) $f(x) = \sqrt{\frac{x-1}{x+2}}$ $\begin{cases} x \neq 0 \\ x \neq -2 \end{cases}$ $D = \{x \in \mathbb{R} / x \neq 0\}$

$f'(x) = \sqrt{\frac{x-1}{x+2}}$ $\frac{-(x-1)}{x+2} > 0$ $\begin{cases} x \neq -2 \\ x \neq 0 \end{cases}$

$D = \{x \in \mathbb{R} / x > 1\}$

$f)$ $f(x) = \sqrt{\frac{x-4}{x-1}}$

$\frac{x-4}{x-1} \geq 0$

$\frac{x-4}{x-1} \geq x - 4$

$\frac{x-4}{x-1} - x \geq 0$

$\frac{x-4-x^2+4x}{x-1} \geq 0$

$\frac{-x^2+5x-4}{x-1} \geq 0$

$-x^2+5x-4 = 0$

$x_1 = 1$

$x_2 = 4$

$x \neq 1$

$x \neq 1$

$\frac{x}{x} \geq x - 2$

$\frac{x}{x} - x \geq 0$

$\frac{x-x^2}{x} \geq 0$

$\frac{-x^2+x}{x} \geq 0$

$-x^2 + x = 0$

$x(-x+1) = 0$

$x_1 = 0$

$x_2 = 1$

$x \neq 0$

$x \neq 0$

$\frac{x-4}{x-1} \geq 0$

$\frac{x-4}{x-1} \geq x - 4$

$\frac{x-4}{x-1} - x \geq 0$

$\frac{x-4-x^2+4x}{x-1} \geq 0$

$\frac{-x^2+5x-4}{x-1} \geq 0$

$-x^2 + 5x - 4 = 0$

$x_1 = 1$

$x_2 = 4$

$x \neq 1$

$x \neq 1$

$S_1 \cap S_2 = \{x \in \mathbb{R} / x \in \mathbb{R}, x \neq 1\}$

EXM 4) $\frac{(x-1)}{(x+2)} \geq \frac{1}{2}$

$\frac{(x-1)}{(x+2)} - \frac{1}{2} \geq 0$

$\frac{2(x-1) - (x+2)}{2(x+2)} \geq 0$

$\frac{2x-2-x-2}{2(x+2)} \geq 0$

$\frac{x-4}{2(x+2)} \geq 0$

$x - 4 = 0$

$x = 4$

$2(x+2) = 0$

$x = -2$

$x \neq -2$

$S_1 \cap S_2 = \{x \in \mathbb{R} / x \in \mathbb{R}, x \neq -2\}$

EXM 5) $f(x) = \begin{cases} 3x+2 & \text{se } x \geq 0 \\ -3x+2 & \text{se } x < 0 \end{cases}$ $g(x) = 3x - 2$

$g \circ f(x) = ?$

$f \circ g(x) = ?$

$f \circ g(x) = ?$

$f(x) = \begin{cases} 3x+2 & \text{se } x \geq 0 \\ -3x+2 & \text{se } x < 0 \end{cases}$

$f(g(x)) = \begin{cases} f(3x-2) & \text{se } x \geq 0 \\ f(3x-2) & \text{se } x < 0 \end{cases}$

$f(g(x)) = \begin{cases} 3(3x-2)+2 & \text{se } x \geq 0 \\ 3(3x-2)+2 & \text{se } x < 0 \end{cases}$

$\boxed{g \circ f(x) = \begin{cases} 9x-4 & \text{se } x \geq 0 \\ -9x+4 & \text{se } x < 0 \end{cases}}$

$f(g(x)) = \begin{cases} f(3x-2) & \text{se } 3x-2 \geq 0 \\ f(3x-2) & \text{se } 3x-2 < 0 \end{cases}$

$f(g(x)) = \begin{cases} 3x-2 & \text{se } x \geq 0 \\ -3x+2 & \text{se } x < 0 \end{cases}$

$\boxed{f \circ g(x) = \begin{cases} 3x-2 & \text{se } x \geq 0 \\ -3x+2 & \text{se } x < 0 \end{cases}}$

EXM 6) $f(x) = x^2$

$f(x) = x^2$

$f(x) = x^2$

GRÁFICO DE f EM AZUL

FUNÇÃO PAR

FUNÇÃO ÍMPAR

EXM 7) $f(x) = \sqrt{\frac{x}{x-1}}$ $\begin{cases} x \neq 1 \\ x \geq 0 \end{cases}$

$x^2 + x \neq 0$

$x^2 + x = 0$

$x \neq 0$

$x(x+1) = 0$

$x_1 = 0$

$x_2 = -1$

$x \neq 0$

$x \neq -1$

$D = \{x \in \mathbb{R} / x \neq 0, x \neq -1\}$

$f \circ g = \sqrt{\frac{x}{x-1}}$

$x^2 + x \neq 0$

$x^2 + x = 0$

$x(x+1) = 0$

$x_1 = 0$

$x_2 = -1$

$x \neq 0$

$x \neq -1$

$x = 1$

$\frac{x}{x-1} \geq 0$

$\frac{x}{x-1} \geq 0$

$\frac{x}{x-1} \geq 0